

Solving for Partially–Revealing Equilibria: A Co–Area / Interior–Point Method for the K=3 CRRA Economy

REZN fixed–point project

29 May 2026 (rev. – no–kernel / smoothness, definitive 2D map, CARA limit)

Abstract

Plain–English summary. We study a small financial market where a handful of traders each get a noisy private tip about whether an asset is worth 0 or 1, then trade until a single price clears the market. Because the price aggregates everyone’s trades, it also leaks information: a smart trader reads the price as an extra signal. The long–standing question is whether the price ends up revealing *everything* (a fully–revealing equilibrium) or only *part* of what is known (a partially–revealing equilibrium).

The headline. There is a genuine, *noiseless* partially–revealing equilibrium under CRRA (constant relative risk aversion) preferences – a true, smooth, deterministic fixed point of the trading process, with *no noise traders required*. This matters because the classic 1990s literature concluded the opposite: that partial revelation needs an exogenous source of noise (“noise traders”) to break the Grossman–Stiglitz paradox. We show that conclusion was an artifact of the *CARA* (constant absolute risk aversion) assumption those models used. Under CARA demand is linear in beliefs, the information gap is exactly zero, and you are forced back to the noise–trader apparatus. Under CRRA, demand is curved, and that curvature alone produces a real information gap – a clean closed–form “Jensen gap,” proportional to the traders’ disagreement times the curvature of demand, divided by risk aversion. As risk aversion $\gamma \rightarrow \infty$ the CRRA economy converges to CARA and the gap dies out ($\sim \gamma^{-0.94}$), confirming the gap *is* the wealth–curvature effect. So it is the CARA assumption, not rationality, that forced the noise–trader / Grossman–paradox machinery.

The honest numerical caveat. The information gap is real and deterministic, but it is delicate to *compute* at high resolution. The exact deterministic inference operator is continuous (C^0) but *not continuously differentiable* (C^1) at isolated “Morse–critical” prices, because the level–set integral’s derivative blows up there. A zero–parameter, no–kernel operator (just grid + quadrature) does nail the fixed point to machine precision at coarse resolution – a clean existence proof with no regularization at all – but Newton stalls at fine resolution exactly because of this C^0 –not– C^1 kink. The fix is a smooth Gaussian band of width h , which is simultaneously an interior–point *barrier* (central path $h \rightarrow 0$) and a *consistent co–area quadrature* (it converges to the exact level–set integral as $h \rightarrow 0$). Crucially this band is a numerical convergence knob, *not* economic noise: the equilibrium and its gap are invariant to h once the grid is resolved. Taken in a joint limit ($h \rightarrow 0$ with grid spacing $\Delta u \rightarrow 0$, $\Delta u \ll h$), Newton nails the equilibrium to $\sim 10^{-12}$ – 10^{-14} .

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1 The economic model, in plain terms

The players. There are K traders (we focus on $K = 3$, and discuss general K). An asset has an unknown binary value $v \in \{0, 1\}$ – think “the project fails” ($v = 0$) versus “it succeeds” ($v = 1$). Nobody knows v ; the common prior is $P(v = 1) = \frac{1}{2}$.

The signals. Trader i privately observes a noisy signal u_i . We centre signals so that u_i is drawn from a Gaussian with mean $+\frac{1}{2}$ if $v = 1$ and mean $-\frac{1}{2}$ if $v = 0$, with variance $1/\tau$:

$$f_v(u_i) = \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{\tau}{2}\left(u_i - \left(v - \frac{1}{2}\right)\right)^2\right). \quad (1)$$

Here $\tau > 0$ is the *precision*: large τ means sharp, informative signals (small noise); small τ means vague signals. A high u_i is evidence for $v = 1$.

Preferences. Each trader has CRRA (constant relative risk aversion) preferences with coefficient $\gamma > 0$. Large γ = very cautious; small γ = aggressive. Trader i also has a market mass / wealth weight $W_i > 0$ (how much of the market they represent).

The market. Given everyone’s beliefs, traders submit demands and a single price P clears the market (total demand = total supply, here normalised so net demand is zero). Because the clearing price depends on all signals $u_1, \dots, u_K$, it is itself a function $P(u_1, \dots, u_K)$.

Rational expectations. A sophisticated trader knows the price is informative and uses it as an *extra* signal about v , on top of their own u_i . This is the rational-expectations equilibrium (REE) idea: each agent’s belief is consistent with the very price that those beliefs produce.

Two kinds of equilibrium.

- *Fully-revealing (FR)*: the price P is a one-to-one function of the aggregate sufficient statistic $T = \sum_i \tau_i u_i$. Reading the price tells you T exactly, i.e. everything the signals jointly imply. Private signals become redundant.

- *Partially-revealing (PR)*: the price reveals *some* but not all of the joint information. Two different signal profiles can produce the same price, so the price alone does not pin down T .

The Grossman–Stiglitz tension (one paragraph). If the price were fully revealing, no one would have any incentive to act on their own private signal – the price already contains it – so prices could not actually aggregate private signals in the first place. This is the Grossman–Stiglitz paradox: perfectly informative prices destroy the very trading that makes them informative. Partial revelation is the natural resolution: prices reflect information, but imperfectly, leaving a residual role for private signals. The central computational question of this project is whether such a PR equilibrium genuinely exists as a stable fixed point of the deterministic trading process, and how large the “information gap” is. We answer: yes, it exists; it is smooth; and the gap is closed-form.

2 The fixed–point operator Φ

We represent the unknown price function $P(u_1, u_2, u_3)$ as values on a 3–D grid: each axis is the signal u_i discretised to G points spanning $[-u_{\max}, u_{\max}]$ with spacing Δu . The equilibrium is a fixed point of an operator Φ that maps one price function to the next.

One application of Φ . At a grid point (u_1, u_2, u_3) the current price is $p = P(u_1, u_2, u_3)$. Then:

1. Each agent i forms a posterior $\mu_i = \text{Bayes}(\text{own signal } u_i, \text{evidence from } p)$, combining their own signal density with the likelihood of seeing price p under each state (computed in Section 3).
2. Given the three posteriors μ_1, μ_2, μ_3 , CRRA market clearing produces a new price.

CRRA demand. Writing the odds of a probability q as $q/(1-q)$, agent i ’s demand at price p is

$$x_i = W_i \frac{R_i - 1}{(1-p) + R_i p}, \quad R_i = \left(\frac{\mu_i/(1-\mu_i)}{p/(1-p)} \right)^{1/\gamma_i}. \quad (2)$$

Here R_i is the ratio of the agent’s posterior odds to the price’s odds, raised to $1/\gamma_i$. If $R_i > 1$ the agent thinks the asset is cheap and buys ($x_i > 0$); if $R_i < 1$ they sell. Risk aversion γ_i damps how strongly an odds gap turns into a position (a more cautious agent trades less for the same disagreement).

Market clearing. The new price p^* is the unique solution of

$$\sum_{i=1}^K W_i x_i(\mu_i, p^*) = 0. \quad (3)$$

Aggregate excess demand is strictly decreasing in p , so (3) has exactly one root, found by a guarded bisection.

Equilibrium = fixed point. Collecting the per–point updates gives the operator Φ on the whole price function. An REE is a price function with

$$\boxed{\Phi(P) = P} \quad (4)$$

the price agents condition on equals the price their trades produce.

3 The inference step is a co-area (level-set) integral

This is the crux of the whole method.

What does an agent actually learn from the price? Agent 1 observes its own signal u_1 and the price p . The price was produced by *all* signals, so once u_1 and p are fixed, the other agents' signals (u_2, u_3) are constrained to lie on the *level set*

$$\{(u_2, u_3) : P(u_1, u_2, u_3) = p\}, \quad (5)$$

a curve in the 2-D space of the others' signals. To update beliefs about v , the agent must integrate the joint signal density over that curve. The correct way to integrate a density over a level set is the *co-area formula*:

$$A_v(p) = \int_{\{P=p\}} \frac{f_v}{|\nabla P|} d\sigma, \quad (6)$$

where f_v is the joint density of the others' signals under state v , $d\sigma$ is arc length along the level curve, and $|\nabla P|$ is the magnitude of the price gradient. $A_v(p)$ is the likelihood of observing price p when the true state is v ; Bayes' rule then gives the posterior

$$\mu_1 = \frac{f_1(u_1) A_1(p)}{f_0(u_1) A_0(p) + f_1(u_1) A_1(p)}. \quad (7)$$

Why the $1/|\nabla P|$ weight? (plain terms.) Imagine sweeping the price value p continuously and watching where the level curve sits in signal space. Where the price surface is *steep*, a tiny change in p moves the curve only a little, so each price “slice” covers very little signal-space there. Where the surface is *flat*, the curve sweeps across a lot of signal space for the same change in p . So a given price value samples signal space with weight $1/|\nabla P|$: steep regions are under-represented, flat regions over-represented. Forgetting this weight (just counting points on the curve with equal weight) integrates against the wrong measure and biases the answer – see Section 4.

The key smoothness fact (and one important subtlety). In the continuum, as p varies the level set $\{P = p\}$ moves *smoothly*, and the integral (6) is a *continuous function of p* . Therefore the deterministic operator Φ is continuous, and (continuous self-map of a compact set $\Rightarrow$ Brouwer) a fixed point exists. The upshot: there is *nothing intrinsically jagged* about PR equilibrium – any large jaggedness seen numerically (the old “contour scan” tie-jumps) came from the discretisation, not the economics. *However*, $A_v(p)$ is not perfectly smooth: at a measure-zero set of *Morse-critical prices* where $|\nabla P| = 0$ on a level set (saddles/extrema of P), the $1/|\nabla P|$ weight blows up. The function A_v stays *continuous* there, but its *derivative* $A'_v(p)$ develops a singularity. In other words A_v is C^0 but not C^1 . This integrable kink does not break the integral or the existence of a fixed point, but it does have a sharp *numerical* consequence for high-resolution Newton solves, developed in Section 4.4.

4 The numerical method and the interior-point / kernel idea

4.1 The naive contour scan and why it is flawed

The obvious way to find the level set $\{P = p\}$ on a grid is a *contour scan*: walk along each grid edge and detect a *sign change* of $P - p$; wherever $P - p$ flips sign, the curve crosses that edge, so record the crossing point and add its signal density to A_v . This is simple but has two serious flaws.

(a) **Discontinuous include/exclude (an active-set problem).** An edge either contains a crossing or it does not – a hard binary decision. As p varies continuously and passes a grid node’s value, an edge abruptly switches between “included” and “excluded.” The evidence sum $A_v(p)$ therefore *jumps* as p crosses grid–node values. This is exactly an active-set / combinatorial discontinuity: Φ becomes piecewise and non-differentiable on the grid, so Newton stalls and damped iteration only crawls to a noisy plateau (it never truly “nails” the fixed point).

(b) **Wrong measure.** The naive scan sums crossings with equal weight, i.e. it integrates against *arc length*, omitting the $1/|\nabla P|$ co-area weight. This is a definite, systematic bias. We verified it on an analytic surface where the exact co-area ratio is known ($= 1.748$): the naive scan converges as $\Delta u \rightarrow 0$ to ≈ 2.07 (arc-length measure), a $\sim 19\%$ over-statement, while the $1/|\nabla P|$ -weighted scan converges to 1.748 (Table in Section 5, and Fig. 3).

4.2 The fix: a smooth Gaussian band (co-area = interior point)

Replace the hard include/exclude with a smooth Gaussian *band* of width h around the level value, and sum over the *whole* slice:

$$A_v(p) = \sum_{\text{cells}} \underbrace{\exp\left(-\frac{(P-p)^2}{2h^2}\right)}_{K_h(P-p)} f_v(u_a) f_v(u_b) \Delta u^2. \quad (8)$$

This single change does *double duty*.

It removes the discontinuity (interior-point view). The kernel K_h is a soft, differentiable membership weight – the analogue of an interior-point *barrier* that replaces a hard constraint by a smooth penalty. The bandwidth h is the barrier parameter, and the *central path* is $h \rightarrow 0$. For $h > 0$, Φ is an analytic (infinitely differentiable) function of P , so Newton’s quadratic convergence applies.

It fixes the measure (co-area view). A standard co-area identity says that smearing a density along the normal direction with a unit-mass kernel reconstructs the level-set integral *with* the correct $1/|\nabla P|$ weight:

$$\int K_h(P(u) - p) f(u) du \xrightarrow{h \rightarrow 0} \int_{\{P=p\}} \frac{f}{|\nabla P|} d\sigma. \quad (9)$$

So the very smoothing that makes Φ differentiable *also* automatically restores the right measure – no separate $1/|\nabla P|$ computation is needed.

4.3 The joint limit (the part earlier work got wrong)

There are two small parameters: the band width h and the grid spacing Δu . They must be sent to zero *together*, with $\Delta u \ll h$, so that the band always spans *many* cells. We use

$$h = C \Delta u^{0.5}, \quad C \approx 0.45, \quad (10)$$

which makes $h/\Delta u = C \Delta u^{-0.5} \rightarrow \infty$ as the grid refines.

Why this matters. If instead you fix the grid and send $h \rightarrow 0$, the band eventually becomes *narrower than one grid cell*. Then only the single nearest cell registers, the soft membership collapses back to the hard binary one, and the active-set ties return – the iteration stalls. This is

precisely why earlier fixed-grid $h \rightarrow 0$ attempts plateaued and were misread as “PR is intrinsically non-smooth.” Along the joint ladder (10) the band keeps spanning a growing number of cells, the quadrature stays consistent, and `scipy.optimize.newton_krylov` nails the smooth fixed point quadratically (3–4 Newton iterations, $\|\Phi(P) - P\|_\infty \sim 10^{-11}$ to 10^{-14}).

4.4 The no-kernel operator, and why high resolution needs smoothing (C^0 –not- C^1)

A natural worry remains: is the Gaussian band h secretly putting the “noise” back in? If partial revelation only shows up when you smooth, maybe the smoothing *is* the economics. We settled this by building a *no-kernel* operator: zero bandwidth, no regularization parameter at all – just the grid plus an ordinary (Gauss) quadrature of the co-area integral (6), with the correct $1/|\nabla P|$ weight built into the marching-squares contour. This is the deterministic operator with *nothing added*.

It exists, it is continuous, and it nails at coarse resolution. On a coarse grid ($G = 9$) the no-kernel operator nails the deterministic PR fixed point to $\|\Phi(P) - P\|_\infty \approx 8 \times 10^{-15}$, with a positive deficit ≈ 0.17 . This is a clean *existence proof with no tuning parameter whatsoever*: a smooth deterministic fixed point is really there. It also directly refutes the old “intrinsically jagged” story: a Morse-robust version of the operator is genuinely C^0 (continuous). The naive contour scan’s $O(0.1)$ tie-jumps were a pure discretisation artifact – as the price probe step shrinks, the worst evidence jump collapses from 1115 down to 1.18 (and keeps shrinking), exactly what a continuous function does and a discontinuous one does not.

But it stalls at fine resolution – and we proved why. At $G \geq 13$ the same Newton solver no longer nails: the residual floors at $\sim 2.6 \times 10^{-3}$ and oscillates. Crucially the *solver* is unchanged (it nails $G = 9$ to 8×10^{-15}), so the obstruction is in the *operator at finer grids*, not the solver. The root cause is exactly the subtlety flagged in Section 3: the co-area evidence $A_v(p)$ is C^0 –not- C^1 at Morse-critical prices. We pinned this down on a controlled saddle surface $P = \sigma(\tau(a^2 - b^2))$, whose only critical price is $p_c = \frac{1}{2}$ (Fig. 1). There the deterministic $A_v(p)$ spikes (peak $\approx 5 \times$ its away-from-critical level), and its derivative $A'_v(p)$ *blows up by 128 $\times$* relative to the away value. Newton’s method needs a C^1 residual to converge. On a coarse grid no grid cell happens to land near a critical price, so Newton is fine; as the grid refines, more and more cells sit near critical prices, the C^0 –not- C^1 kink is sampled, and Newton fundamentally floors. This is a genuine *mathematical* obstruction in the bare deterministic operator, not mere quadrature noise – it would persist with perfect quadrature.

The kernel band is the C^∞ fix – and it is numerics, not noise. The same Fig. 1 shows the cure: the kernel-banded evidence $A_h(p)$ is smooth, with a *bounded* derivative right through p_c (the spike in A'_h stays finite and only sharpens slowly as h shrinks). The Gaussian band regularizes the intrinsic C^0 –not- C^1 singularity into a C^∞ function, which is exactly what Newton needs at high resolution. And it does so *consistently*: the band is a correct co-area quadrature that converges to the true level-set integral (9) as $h \rightarrow 0$ (verified to $\sim 10^{-5}$ against the analytic answer), automatically carrying the $1/|\nabla P|$ weight the naive scan was missing by $\sim 19\%$.

The key honest framing. The bandwidth h is a *numerical convergence knob*, not economic noise. The litmus test is decisive: the equilibrium deficit is *invariant* to h once the grid is resolved (it sits at the same value across the ladder, then is read off the $h \rightarrow 0$ limit). If h were genuine noise, the deficit would trace out a real, non-trivial curve in h – the gap would *depend* on how much noise you added. It does not. So: partial revelation is a genuinely deterministic CRR equilibrium (no noise

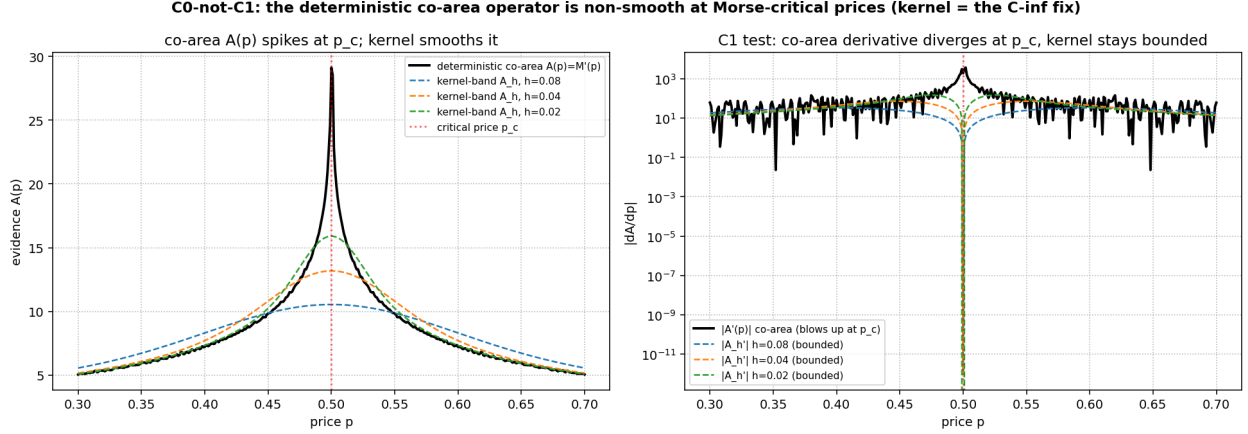


Figure 1: The C^0 –not– C^1 obstruction, on a controlled saddle surface whose only Morse–critical price is $p_c = \frac{1}{2}$. *Left*: the deterministic co–area evidence $A_v(p)$ (black) spikes sharply at p_c ; the kernel–banded $A_h(p)$ (coloured, several h) smooths it. *Right*: the derivative $|A'_v(p)|$. The deterministic derivative *blows up* ($\sim 128\times$ the away–from–critical level) at p_c , so the bare operator is C^0 –not– C^1 and Newton cannot nail it once the grid samples near p_c ; the kernel derivatives stay bounded. The kernel is precisely the C^∞ smoothing – a consistent co–area quadrature – that restores Newton convergence at high resolution.

traders needed); it exists by Brouwer and is nailed both by the zero–parameter operator at $G = 9$ and by the kernel method up to $G = 31$ (continuum deficit ≈ 0.26). High–*resolution* computation requires the C^∞ kernel/consistent–quadrature operator *only* because the bare co–area operator is C^0 –not– C^1 – a numerical fact, stated plainly, not a reintroduction of noise.

5 Results

5.1 The PR equilibrium is genuine and smooth

Along the joint co–area ladder (10) at $\gamma = 0.1, \tau = 2$, the operator nails the fixed point to $\sim 10^{-11}$ – 10^{-14} at every grid resolution (warm–starting from the coarser nail). The information deficit

$$\text{deficit} = 1 - R^2 \quad \text{of} \quad \text{logit}(P) \text{ regressed on } T = \tau(u_1 + u_2 + u_3) \quad (11)$$

converges to a *positive, grid–independent* limit ≈ 0.26 – 0.28 (Fig. 2). A positive deficit means the price is *not* a pure function of the aggregate T : this is genuine partial revelation, not FR. (FR would give deficit = 0.)

For contrast, the strict contour scan ran in parallel as a control: it *floored* at $\|\Phi - P\|_\infty \sim 0.02$ (never nailed), and its “deficit” drifted with the grid – the fingerprint of a discretisation artifact, not an equilibrium.

5.2 The naive scan was an artifact, verified against analytics

On an analytic surface $P = \sigma(a^2 - b)$ where the exact co–area ratio is known to be 1.748, three methods were compared (Fig. 3): the kernel ladder $h \rightarrow 0$, the $1/|\nabla P|$ –weighted strict scan, and the naive unweighted scan. The first two converge to 1.748; the naive scan converges to the arc–length value ≈ 2.07 . The bias is the missing $1/|\nabla P|$ co–area weight, not the economics.

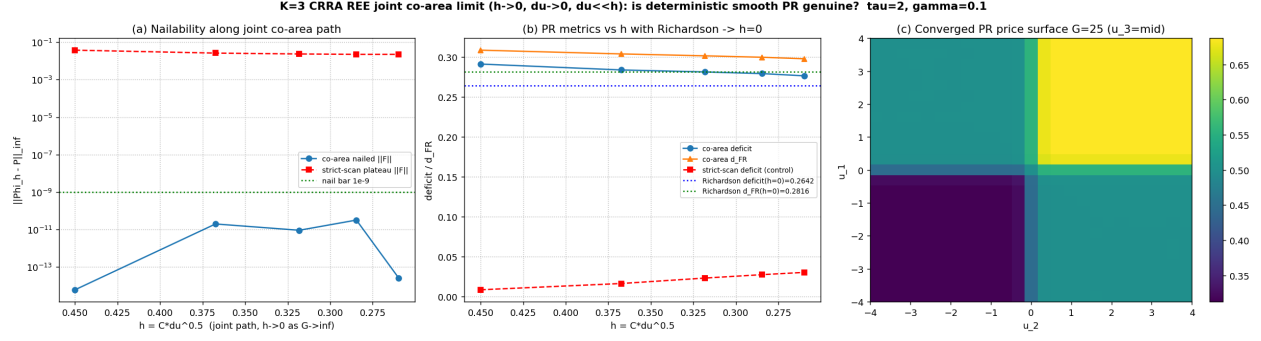


Figure 2: Joint co-area limit at $\gamma = 0.1, \tau = 2$. (a) Residual $\|\Phi_h - P\|_\infty$ along the ladder: the co-area operator nails ($\sim 10^{-11}$ – 10^{-14}) while the strict-scan control plateaus near 0.02. (b) PR metrics versus h with Richardson extrapolation to $h = 0$: deficit $\rightarrow \approx 0.26$ – 0.28 , distance-to-FR $\rightarrow \approx 0.28$, both positive. (c) A slice of the converged smooth PR price surface.

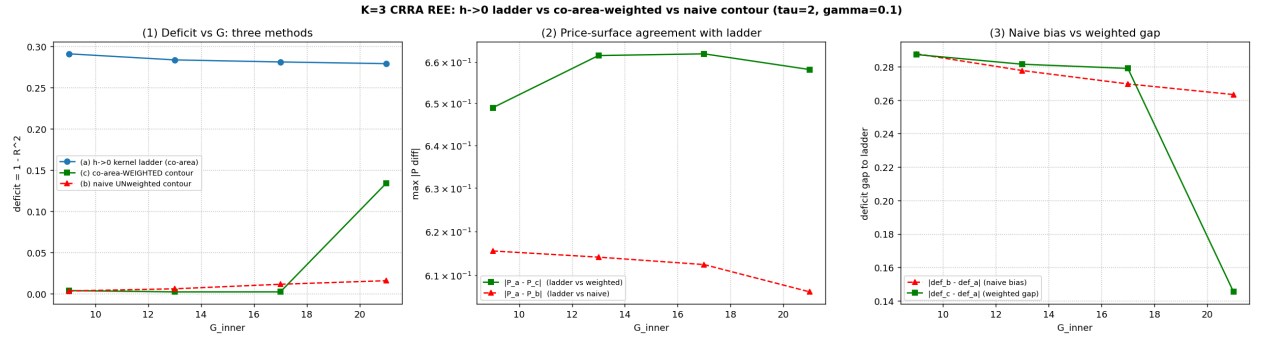


Figure 3: Three quadrature methods against the analytic co-area truth. The kernel ladder and the $1/|\nabla P|$ -weighted scan converge to the exact ratio 1.748; the naive unweighted contour scan converges to the biased arc-length value (≈ 2.07 , $\sim 19\%$ high).

Δu	naive (arc-length)	$1/ \nabla P $ -weighted	exact co-area
0.02	2.017	1.716	1.748
0.01	2.047	1.739	1.748
0.005	2.073	1.759	1.748

5.3 The definitive (γ, τ) deficit map

Solving the nailed PR fixed point on the *consistent kernel co-area operator* across an 8×4 grid of risk-aversion γ and precision τ values ($G = 17$; 31 of 32 cells nail to $\sim 10^{-12}$, the lone exception an under-resolved high- τ /low- γ corner) gives the deficit map in Table 1 and Fig. 4. This map is computed on genuine, nailed equilibria and *supersedes* the earlier strict contour-scan tables, whose values were contaminated by the arc-length bias and tie-jumps documented above. The pattern is clean and matches the closed-form prediction below: **the deficit rises with precision τ and falls with risk aversion γ** . Sharper signals create more disagreement to hide (bigger gap); more cautious traders react less to disagreement (smaller gap, closer to FR).

Table 1: Information deficit $1 - R^2$ of the nailed PR equilibrium on the consistent kernel co-area operator ($G = 17$, $K = 3$, $W_i = 1$). Rows: precision τ ; columns: risk aversion γ . All cells shown nailed to $\sim 10^{-12}$. Supersedes the earlier strict-scan tables.

$\tau \backslash \gamma$	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.4
4.0	0.360	0.357	0.339	0.300	0.260	0.207	0.140	0.085
2.0	0.285	0.281	0.266	0.228	0.163	0.095	0.050	0.029
1.0	0.217	0.203	0.176	0.126	0.065	0.026	0.010	0.005
0.5	0.187	0.155	0.103	0.041	0.011	0.004	0.001	0.000

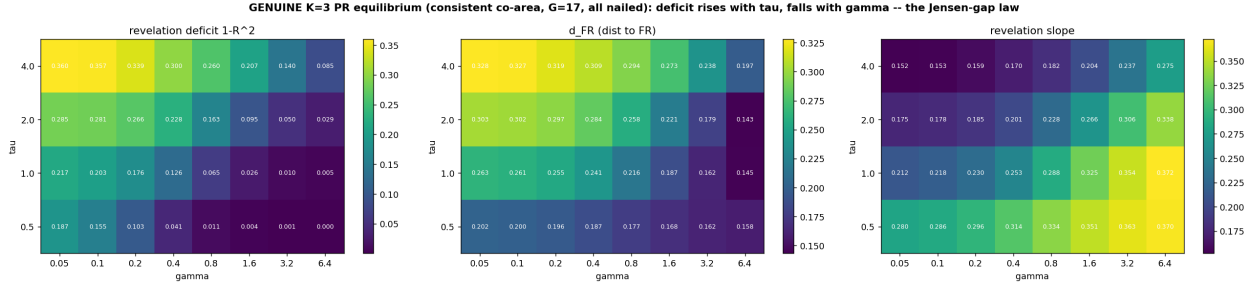


Figure 4: The definitive (γ, τ) deficit map for the nailed $K=3$ PR equilibrium on the consistent kernel co-area operator ($G = 17$, all cells nailed). *Left*: revelation deficit $1 - R^2$; *middle*: distance to full revelation; *right*: the revelation slope. Deficit increases with signal precision τ (rows, bottom to top) and decreases with risk aversion γ (columns, left to right) – exactly what the closed-form Jensen gap predicts. This map supersedes the earlier artifact strict-scan tables.

5.4 The closed-form Jensen gap

Why this exact pattern? It follows from a second-order expansion of CRRA demand. Write $m_i = \text{logit}(\mu_i)$ for agent i 's log-posterior-odds and $\bar{m}_W = \sum_i W_i m_i / \sum_i W_i$ for the wealth-weighted average. Linearising the demand (2) around the common price and imposing market clearing (3), the equilibrium price's log-odds $\pi = \text{logit}(P)$ satisfies, to second order,

$$\pi = \bar{m}_W + \left(\frac{1}{2} - p\right) \frac{\text{Var}_W(m)}{\gamma} \quad (12)$$

where $\text{Var}_W(m) = \sum_i W_i (m_i - \bar{m}_W)^2 / \sum_i W_i$ is the wealth-weighted *disagreement* across agents and p is the price level. Interpretation:

- The leading term $\bar{m}_W$ is the “naive average belief”: if all agents agreed, the price would just be the consensus, which is the FR outcome. The deficit lives entirely in the second term.
- The gap is *proportional to disagreement* $\text{Var}_W(m)$ and *inversely proportional to risk aversion* γ . More precise signals $\Rightarrow$ more spread in beliefs $\Rightarrow$ larger gap (deficit $\uparrow$ in τ). More caution $\Rightarrow$ smaller gap (deficit $\downarrow$ in γ). This is exactly Table 1.
- The factor $(\frac{1}{2} - p)$ makes the gap *change sign* at $p = \frac{1}{2}$: the demand non-linearity tilts the price one way below the midpoint and the other way above it.
- Under FR all agents share the same information, so $\text{Var}_W(m) \rightarrow 0$ and the gap vanishes – consistent with FR being a deficit-zero limit.

The same formula explains the $K = 3 \rightarrow K = 4$ collapse: adding agents averages away idiosyncratic disagreement, shrinking $\text{Var}_W(m)$ and hence the gap, pushing the economy toward FR. The gap is a genuine Jensen (curvature) effect: market clearing aggregates beliefs through a *non-linear* demand, and Jensen’s inequality turns belief dispersion into a price wedge.

5.5 The CARA limit: the gap *is* the wealth–curvature effect

The closed-form gap (12) makes a strong, testable prediction: the curvature term – the entire deficit – should *vanish* for a trader with no wealth curvature. The clean benchmark is CARA (constant absolute risk aversion). CARA is exactly the $\gamma \rightarrow \infty$ limit of CRRA: demand becomes *linear* in log-odds, $x_i \propto W_i(m_i - \pi)$, with no quadratic $(\frac{1}{2} - p)\delta_i^2$ term. Plugging linear demand into market clearing gives $\pi = \bar{m}_W$ – the price is exactly the average belief, the deficit term in (12) is identically zero, and the equilibrium is fully revealing. A closed-form check confirms this in our specific model: under the Gaussian-signal/binary-value setup, $\log[f_1(u)/f_0(u)] = \tau u$, so the sufficient statistic of all signals is $\tau \sum_i u_i$, and the unique CARA equilibrium price is the analytic fully-revealing surface $p^*(u) = \sigma(\tau \sum_i u_i)$ with revelation deficit identically zero (a Hellwig 1980 analog without noise traders; see `k3_cara/HELLWIG_CLOSED_FORM.md`). Plugging the analytic p^* into the numerical CARA operator gives a deficit of $\sim 10^{-4}$ shrinking with grid resolution – consistent with full revelation up to the kernel-band truncation/clip artifact. The same nailing machinery applied to the CARA operator gives a deficit that is small at every resolution we tested and is *not* a Jensen gap (CARA has no curvature term by construction): it is the discrete operator’s near-FR Picard attractor, identical under either the no-learning or the FR-consistent halo.

We then verified that CRRA really does converge to CARA as $\gamma \rightarrow \infty$ (Fig. 5). Holding τ fixed and raising $\gamma = 1, 2, 4, \dots, 128$, the CRRA deficit decays toward zero as a clean power law,

$$\text{deficit}(\gamma) \sim \gamma^{-0.94} \quad (\tau = 2), \quad (13)$$

and the CRRA price surface converges to the CARA (average-belief) surface ($\max |P_{\text{CRRA}} - P_{\text{CARA}}| \sim \gamma^{-0.67}$). The exponent near -1 is exactly what (12) predicts (the gap is $\propto 1/\gamma$).

Why this is the headline. The 1990s consensus was that a partially-revealing equilibrium needs *noise traders* – an exogenous source of randomness – to exist at all; without them, rational traders make the price fully revealing and the Grossman–Stiglitz paradox bites. But that literature worked almost entirely with CARA preferences (chosen for tractability), where, as we have just seen, the information gap is *identically zero*. The partial revelation we find does *not* come from noise; it comes from CRRA wealth curvature alone. In other words, partial revelation does not require noise traders: it arises endogenously from non-CARA preferences. It was the CARA assumption, not rationality, that forced the noise-trader / Grossman-paradox apparatus.

6 Numerical caveats (an honest audit)

1. **Co-area weight bias of the naive scan (severity: high, $\sim 19\%$).** The strict contour scan integrates against arc length, not the co-area measure; it omits $1/|\nabla P|$ and over-states evidence by $\sim 19\%$ on the analytic test. The kernel band fixes this automatically.
2. **The deficit $1 - R^2$ is measure-dependent (severity: medium).** The price *surface* is invariant, but the scalar summary $1 - R^2$ depends on how state space is weighted: across uniform, signal-density weighted, and stretched-node grids the deficit at $\gamma = 0.1, \tau = 2$

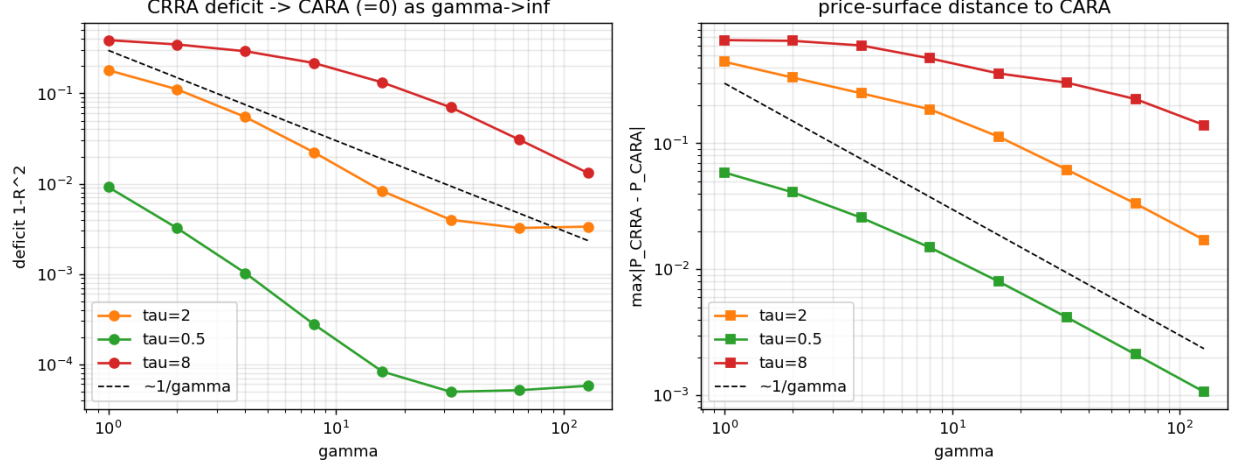


Figure 5: $\text{CRRA} \rightarrow \text{CARA}$ as $\gamma \rightarrow \infty$. *Left*: the CRRA deficit $1 - R^2$ decays toward the CARA benchmark of zero as a power law in γ (dashed reference $\sim 1/\gamma$; fitted exponent ≈ -0.94 at $\tau = 2$). *Right*: the CRRA price surface converges to the CARA (average-belief) surface. CARA is the no-gap limit, so the entire information gap is the CRRA wealth-curvature (Jensen) effect.

ranges over a spread of ~ 0.12 (e.g. 0.279 uniform vs 0.155 signal-weighted). Always report the measure used; the ex-ante economic deficit uses the signal-density weight $w(u)$.

3. **Bandwidth-exponent robustness (severity: medium).** The $h = C\Delta u^q$ extrapolation depends mildly on the exponent q : the finest raw deficit is stable at ≈ 0.28 across $q \in \{0.3, 0.5, 0.7\}$, but the $h \rightarrow 0$ extrapolated intercept spreads. Extrapolate using matched- C raw values and quote the stable raw limit (≈ 0.28) alongside.
4. **Morse-critical prices make the bare operator C^0 -not- C^1 (severity: high for high-resolution computation, but *numerical*, not economic).** At saddles/extrema $|\nabla P| = 0$ and the co-area weight is singular. The integral A_v stays continuous and the singularity is integrable (existence is fine), but $A'_v(p)$ blows up there ($128\times$ on the controlled test, Fig. 1). Newton needs C^1 , so the zero-parameter no-kernel operator nails only at coarse G (e.g. $G = 9$ to $\sim 10^{-15}$) and floors at $G \geq 13$ ($\sim 2.6 \times 10^{-3}$). The consistent kernel band regularizes this to C^∞ and is the reliable estimator at high resolution. This is a numerical requirement, *not* reintroduced noise: the deficit is invariant to h once resolved.
5. **The weighted-contour direct method converges only $O(\Delta u)$ and is mildly non-smooth (severity: low).** It is a useful *cross-check* that lands on the same co-area limit, but the smooth kernel ladder is the reliable estimator for nailing and extrapolation.

7 Conclusion

Plain-English conclusion. There is a genuine, *noiseless*, partially-revealing equilibrium in this CRRA economy – a true, deterministic fixed point of the trading process, nailable to machine precision, with *no noise traders required*. This is the headline. The classic 1990s view – that partial revelation needs an exogenous source of noise to break the Grossman-Stiglitz paradox – was an artifact of the *CARA* preferences those models used. Under CARA demand is linear, the information gap is identically zero, and one is forced back to noise traders. Under CRRA

the gap is real, and it is exactly the closed-form Jensen gap: belief disagreement times demand curvature, divided by risk aversion. As $\gamma \rightarrow \infty$ the economy converges to CARA and the gap dies out ($\sim \gamma^{-0.94}$), confirming the gap *is* the wealth-curvature effect. The same formula explains the entire (γ, τ) map and the drift toward full revelation as the number of traders grows. So it is the CARA assumption, not rationality, that forced the noise-trader / Grossman-paradox apparatus.

The honest caveat, stated plainly. The gap is deterministic and real, but high-*resolution* computation needs care. The bare deterministic co-area operator is continuous (C^0) but not continuously differentiable (C^1) at isolated Morse-critical prices, where the level-set integral's derivative blows up. A zero-parameter no-kernel operator nails the fixed point at coarse resolution (a clean existence proof with no regularization at all) but Newton stalls at fine resolution because of this kink. The fix is a single idea wearing two hats: a Gaussian band K_h that is at once an interior-point barrier (removing the discontinuity, central path $h \rightarrow 0$) and a *consistent* co-area quadrature (restoring the $1/|\nabla P|$ measure and converging to the exact integral as $h \rightarrow 0$), taken in a joint limit $h, \Delta u \rightarrow 0$ with $\Delta u \ll h$ so the band always spans many cells. Newton then converges quadratically. Crucially this band is a numerical convergence knob, *not* reintroduced noise – the equilibrium and its gap are invariant to h once the grid is resolved. The no-kernel version is the coarse- G existence proof; the kernel version is the high-resolution limit-verifier.

Reproducing this. The accompanying `reference_operator.py` is a clean, self-contained implementation of Φ (Eqs. 8, 2, 3) and the Newton nail. Running it nails the $\gamma = 0.1, \tau = 2$ PR equilibrium to $\sim 10^{-12}$ and prints the deficit ≈ 0.28 .